



Society of Petroleum Engineers

**SPE-229004-MS**

## **Game-Changing Technology in Downhole Monitoring: The Frequency Division Multiple Access Telemetry Enables the PDHMS to Break the Noise Barrier in ESP Environments**

A. J. Abdulghani, Aramco, Saudi Arabia; O. A. Ogunsanwo, SLB, Saudi Arabia; A. R. Ssafwany and H. A. Muailu, Aramco, Saudi Arabia; S. A. Al-Namlah and C. J. Benjamin, SLB, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229004-MS](https://doi.org/10.2118/229004-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

### **Abstract**

Reliable downhole monitoring is key for optimizing production, especially in wells equipped with Electric Submersible Pumps (ESPs). Conventional Permanent Downhole Monitoring Systems (PDHMS) often face challenges because of electromagnetic interference from ESPs, leading to signal degradation. This paper introduces an innovative PDHMS that uses DC power and Frequency Division Multiple Access (FDMA) telemetry to address these challenges. The aim is to evaluate its concept and performance in challenging environments, demonstrating its ability to provide high-fidelity data acquisition through innovative frequency management.

The new PDHMS employs FDMA telemetry, allocating unique frequency bands to each sensor, enabling simultaneous communication without interference. This system also pioneers the use of DC power, offering enhanced stability and efficiency. The ability to adjust frequencies dynamically ensures uninterrupted data transmission, even in the presence of ESP noise. Field trials were conducted in high-noise environments to assess the system's performance, focusing on data accuracy, sensor differentiation, and noise immunity.

The results confirmed the robustness and long-term reliability of the new PDHMS in challenging downhole environments. FDMA-based telemetry reliably allocated unique frequency bands to each gauge, enabling simultaneous interference-free data transmission. During ESP operation, the system showed full immunity to electromagnetic noise exceeding 40 dB, preserving data integrity throughout transient pressure and temperature events. The system delivered near-real-time monitoring with sub-second latency ( $<0.8$  s) and accurately distinguished up to 16 sensors with 99.7% accuracy. Pressure and temperature measurements exhibited exceptional stability and resolution, capturing variations as fine as  $\pm 0.006$  psi and  $\pm 0.005$  °C, demonstrating the system's suitability for high-resolution reservoir surveillance in ESP-equipped wells.

The new PDHMS represents a significant advancement in downhole monitoring technology, particularly in environments dominated by ESP noise. By leveraging an FDMA-based telemetry, the system achieves unparalleled data accuracy, sensor differentiation, and noise immunity. This innovation empowers operators to optimize production, reduce uncertainty, and improve reservoir management, potentially revolutionizing monitoring practices in the industry.

The novelty of this PDHMS lies in its pioneering use of FDMA telemetry. The FDMA technology's ability to allocate individual frequencies and dynamically adjust to avoid ESP noise is a key innovation, providing a robust solution for real-time monitoring in noisy environments. This combination sets a new standard for data accuracy and reliability in downhole applications.

## Introduction

Continuous and accurate downhole monitoring plays a vital role in optimizing production and safeguarding integrity across the lifecycle of a well. Key reservoir and production parameters—such as pressure, temperature, and fluid composition—enable informed decisions that influence reservoir management, artificial lift performance, and early anomaly detection. Among artificial lift systems, Electric Submersible Pumps (ESPs) are widely adopted for high-rate wells, yet their operation poses a significant challenge to telemetry systems because of the high levels of Electromagnetic Interference (EMI) they generate.

Permanent Downhole Monitoring Systems (PDHMS) have become an essential technology in modern field developments, enabling real-time surveillance and reduced dependence on well interventions. However, the use of conventional PDHMS (e.g. Frequency Shift Keying [FSK]) often leads to performance degradation in ESP-equipped wells. EMI not only disrupts data communication but may also cause partial or complete data loss, which undermines reservoir surveillance programs and compromises decision-making.

To address this critical limitation, this paper presents a novel PDHMS solution that leverages Frequency Division Multiple Access (FDMA) telemetry. By assigning distinct frequency bands to each downhole sensor and using EMI-immune DC power, the system achieves robust and uninterrupted data transmission, even in high-noise environments.

This paper aims to show the technical capabilities, operational reliability, and performance of this FDMA-based DC-powered PDHMS in real-world applications. This introduction sets the stage for a detailed literature review that traces the evolution of downhole monitoring technologies, compares traditional and emerging telemetry methods, and contextualizes the significance of this advancement.

## Literature Review

### The Importance of Downhole Pressure Monitoring

Pressure data is the backbone of petroleum engineering. It is the language that enables reservoir engineers, drilling engineers, and production engineers to communicate and understand wells. After drilling, downhole pressure monitoring and surveillance become effective tools for reservoir and production management. This information supports the characterization of reservoir properties such as permeability, porosity, and fluid contacts, which are essential for building accurate reservoir models. Pressure trends over time also help identify boundary conditions, compartmentalization, and drive mechanisms, all of which influence long-term recovery strategies.

In terms of production optimization, downhole pressure data plays a role in managing artificial lift systems. For example, in wells equipped with ESPs or gas lift systems, monitoring intake and discharge pressures allows for real-time adjustments that enhance lift efficiency and reduce energy consumption. It also enables early detection of performance issues such as pump wear, fluid fallback, or scaling.

From a safety and integrity standpoint, pressure monitoring provides a means to detect anomalies that may show tubing or casing leaks, water breakthrough, or unwanted crossflow between zones. Such events can compromise the structural integrity of the well and pose environmental risks. Timely identification through pressure changes enables corrective action before more severe damage occurs. Maintaining pressures above the bubble point in the phase behavior helps avoid reservoir damage and ensures that production does not shift to the two-phase region and does not exceed fracture gradients, which could lead to formation breakdown or fluid migration.

Moreover, downhole pressure data are essential during Enhanced Oil Recovery (EOR) and stimulation operations. In waterflooding or gas injection schemes, it ensures that injected fluids are effectively distributed throughout the reservoir, promoting uniform sweep and maximizing recovery. Pressure surveillance allows reservoir engineers to detect injection inefficiencies, channeling, or breakthrough early. This feedback supports better reservoir management decisions and improves overall EOR effectiveness.

### **The Evolution of Downhole Pressure Recording**

Various specialized tools and techniques are employed to measure and transmit pressure from the wellbore to the surface, with the choice of method depending on the operational context, desired resolution, and well conditions. Over time, the technology for acquiring this data has evolved significantly, transitioning from rudimentary mechanical gauges to sophisticated, real-time telemetry systems. Early pressure acquisition was primarily achieved through slickline operations, where pressure tools were deployed into the wellbore, left to record data, and retrieved for surface download. While functional and still in use, these methods presented limitations in data availability and operational efficiency, which catalyzed the development of more advanced systems. Furthermore, mechanical gauges were prone to calibration drift, slickline operational challenges and cable wear, limitations in the wellbore drift angle, and the deployment process necessitated production shut-ins, reducing operational efficiency. Despite their simplicity and reliability, the lack of real-time capability and limited resolution constrained their long-term utility.

The advent of wireline technology marked a substantial improvement in downhole pressure recording. In incorporating electrical conductors, wireline tools enabled real-time data transmission to surface systems. Wireline deployment facilitated better depth control and enabled multi-depth surveys in a single run. That said, the requirement for continuous cable deployment posed logistical complications in extended-reach or deviated wells.

In the 1990s, permanent downhole pressure gauges emerged as a transformative solution. These systems, commonly referred to as PDHMS, involved the permanent installation of electronic sensors connected to the surface via electrical cables. PDHMS allowed for continuous, real-time pressure data acquisition without the need to interrupt production operations. This advancement significantly enhanced reservoir surveillance, improved decision-making, and enabled proactive well management. However, traditional electrical PDHMS systems, often reliant on narrow-band telemetry, are vulnerable to EMI, especially in ESP applications, and cable integrity could be compromised in High-Pressure, High-Temperature (HPHT) environments, limiting long-term reliability in certain well conditions.

The operation of ESPs introduces significant EMI, which interferes with downhole sensor communication. Scientists show how EMI disrupts signal transmission, reducing data-fidelity and reliability. Others identify EMI as a leading cause of data transmission failures and note that interference can reduce reading accuracy by up to 30%. Notably, it has been emphasized that the challenges posed by high-frequency noise generated by ESP motor drives, which overlap with telemetry frequencies, limit the effectiveness of conventional PDHMS.

Conventional telemetry methods, such as FSK, often struggle to maintain signal integrity in such noisy environments. These limitations hinder real-time data acquisition and compromise the accuracy of critical parameters, such as pressure, temperature, and flow rate, which are essential for optimizing well performance. As a result, there is a pressing need for innovative solutions that can overcome the barriers posed by EMI while ensuring robust and precise monitoring.

To address these limitations, fiber-optic pressure monitoring systems were introduced in the early 2000s. These systems utilized optical fibers and sensing technologies, such as Distributed Acoustic Sensing (DAS) or Fiber Bragg Gratings (FBG), to measure and transmit pressure and temperature data. Fiber-optic systems offered immunity to EMI and exceptional thermal stability, making them particularly well-suited for wells with electrically noisy environments. Despite this benefit, high installation costs, fiber

termination challenges, complex surface interrogation equipment, and the need for specialized interpretation tools limited the widespread adoption of fiber-optic systems in many fields.

### **Advantages of FDMA Telemetry Over Earlier Methods**

Recent studies have explored advanced telemetry approaches to address EMI, such as FDMA. FDMA is a well-established technique in telecommunications where multiple signals are transmitted simultaneously over different frequency bands. Its application in downhole telemetry is relatively novel but promising. FDMA's primary advantage lies in its ability to provide continuous and parallel data streams from multiple sensors, each assigned to a unique frequency. This reduces the need for time-slot synchronization (as in TDMA) or code orthogonality (as in CDMA), both of which are sensitive to EMI and signal distortion.

These advancements collectively offer a pathway to overcoming the limitations of traditional telemetry and power systems. Features such as dynamic frequency adjustment enable these systems to adapt in real time to varying downhole noise conditions, ensuring uninterrupted data flow and high-fidelity transmission even under harsh conditions. FDMA addresses many of these challenges by using unique frequency bands for each sensor, effectively enhancing signal separation and minimizing EMI. This characteristic makes FDMA inherently more noise-immune, enabling clearer and more reliable data transmission under electrically noisy conditions.

A major limitation of TDMA and similar systems is their reliance on sequential data transmission. As the number of downhole sensors increases, the time delay between transmissions also grows, resulting in latency that hampers real-time decision-making. FDMA resolves this by supporting simultaneous data transmission from multiple sensors. By assigning distinct frequency channels to each device, it eliminates communication bottlenecks and supports real-time monitoring of various well parameters such as pressure, temperature, and flow rate—an essential capability for modern intelligent completions and dynamic reservoir surveillance.

Data integrity and measurement accuracy are also significantly enhanced in FDMA systems. By isolating each sensor's output on a dedicated frequency, FDMA reduces the potential for signal overlap or data corruption—a common issue in shared-bandwidth systems. This isolation results in cleaner signals, fewer retransmissions, and more accurate readings, improving confidence in the data used for reservoir modeling and production optimization.

Scalability is another domain where FDMA outperforms its predecessors. Traditional systems often struggle with congestion when additional sensors are introduced, as all devices must compete for the same communication windows or bandwidth. FDMA, by design, supports easy expansion; new sensors can be added by assigning additional frequency channels, making it ideal for multi-zone monitoring or future field upgrades without affecting existing system performance.

These features make FDMA telemetry particularly reliable in ESP environments. The strong EMI generated by ESP motors often disrupts conventional telemetry channels, leading to frequent signal dropout or corrupted data. FDMA's ability to operate in frequency bands that are less affected by EMI, combined with its resilience through frequency agility, ensures consistent performance even in highly electrified wellbores.

Latency is further minimized in FDMA-based systems. While TDMA introduces delays through its sequential polling mechanism, FDMA's simultaneous transmission model significantly reduces the time required to gather and process downhole data. This low-latency operation is vital for critical real-time applications such as active well control, kick detection, or dynamic lift optimization.

Furthermore, FDMA telemetry is well-suited for integration with advanced downhole architectures, such as intelligent completions and high-density sensor arrays. Its inherent modularity and bandwidth management capabilities make it compatible with emerging technologies, facilitating seamless upgrades and system expansions. This future-proof design ensures that FDMA can evolve alongside the industry's increasing need for high-resolution, high-frequency data acquisition.

## Technology Review of the Installed Novel Gauges with FDMA Telemetry

The new-generation gauges reviewed in this paper are based on quartz sensor technology and can be powered by either DC or AC systems. For this study, the gauges are operated using DC power, as illustrated in Fig. 1a. Their design incorporates dual-seal dry-mate downhole connectors and radial lower tip connectors to ensure proven mechanical robustness in downhole environments.

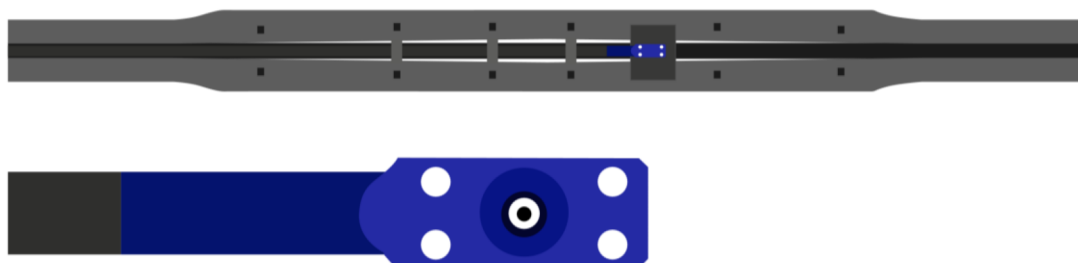


Figure 1a—Illustration of the novel gauges

### Dual-seal Dry-mate Downhole Connector

The dual-seal dry-mate downhole connector features dual redundant metal-to-metal seals with independent verification, providing enhanced reliability. Its design incorporates anti-strain and anti-vibration capabilities to ensure long-term performance. In earlier generations of gauge systems that relied on traditional compression fittings, long-term reliability was often compromised by failures at the cable splice or gauge cable head because of micro-leaks and subsequent fluid ingress. Introducing the dual-seal dry-mate downhole connectors in 2004 effectively addressed this challenge by eliminating the risk of leakage at these critical interfaces, thereby significantly improving the integrity and reliability of permanent downhole monitoring systems. With an extensive field track record, this technology has since been proven to be a robust and dependable solution for long-term reservoir surveillance.

### Radial Lower Tip Connector

The radial lower tip employs a hydraulic gland interface that forms a metal-to-metal seal between the solid gauge mandrel port and the gauge's lower tip. This connector complies with NACE MR0175 standards and has been gas-tested to 18,800 psi and 200 °C. It is validated to design validation grade V0 as per ISO 14998:2013, confirming its suitability for harsh downhole conditions.

### Enhanced Multi-Chip Module (MCM) Technology

The gauges integrate an advanced Multi-Chip Module (MCM) miniaturized electronic circuit, incorporating both semiconductor devices (e.g., transistors and diodes) and passive components (e.g., resistors, inductors, and capacitors). Engineered using high-reliability materials and assembly techniques, the design ensures stable, long-term performance under harsh downhole temperature and pressure conditions.

### Advanced Communication with FDMA

Traditionally, industry-standard gauge telemetry has employed FSK, as illustrated in Fig. 1b. This method functions analogously to a shared telephone line, where each gauge transmits data only when the line is free, introducing several inherent limitations, such as:

- Crosstalk between gauges
- Distance restrictions between gauges
- A limited number of sensors per cable



- Susceptibility to electromagnetic interference

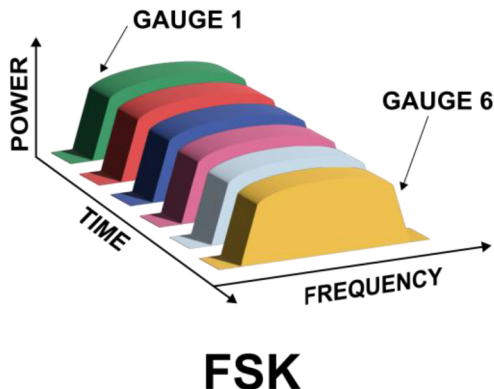


Figure 1b—Showing the FSK protocol of the existing gauges

To overcome these limitations, the novel gauges use telemetry based on FDMA, as shown in Fig. 1c. In this system, each gauge is allocated to a unique frequency band, enabling simultaneous data transmission without interference. The concept can be likened to multiple speakers communicating at once, each using a distinct pitch, making every signal clearly distinguishable. Some of the key technical differentiators of the novel gauge include:

- Bidirectional communication with the surface acquisition unit, fully immune to electromagnetic noise (including ESP environments). This is achieved through automatic frequency reassignment when interference is detected
- Capability to support up to 16 multi-dropped sensors on a single cable while maintaining a 1-second sampling rate
- Qualified for 25+ years of operational life at 200 °C
- Elimination of crosstalk between gauges
- No distance limitations between gauges
- High metrological accuracy leveraging reliable quartz sensor technology

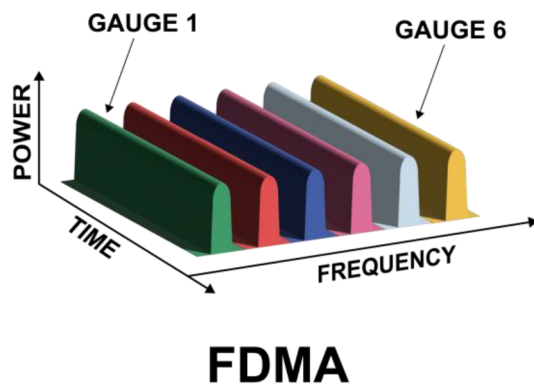


Figure 1c—Showing the FDMA protocol of the novel new gauges

## Engineering Qualification and Validation Testing

The newly developed permanent gauges underwent a comprehensive engineering qualification program and rigorous testing to verify long-term reliability for reservoir and production monitoring. All evaluations

were performed in accordance with Advanced Well Equipment Standards (AWES) and supplemented by additional vendor-specific requirements.

Controlled test cell experiments were conducted to assess accuracy, resolution, and drift stability under simulated downhole pressure and temperature conditions. The gauges were further subjected to repeated power on–off cycling and thermal cycling to replicate harsh and variable operational environments.

To validate mechanical robustness, extensive shock and vibration tests were performed, confirming the gauges' ability to withstand the challenging downhole conditions typical of both production and injection wells. Based on these qualifications, the gauges have been rated for a service life exceeding 20 years at their specified pressure and temperature limits.

A consolidated summary of the qualification and verification tests is provided in [Fig. 2](#), [Table 1](#) and [2](#).

**Table 1—Gauges Validation and Verification Tests Summary**

| Test Description                         | Conditions                                                                 |
|------------------------------------------|----------------------------------------------------------------------------|
| Cold Storage Test                        | Temperature down to –55 °C with various power on/off cycles                |
| Thermal Cycling Test                     | 10 cycles from –25 °C to 210 °C, per AWES 3362-36                          |
| Operating Pressure Cycling Test          | 5 cycles from ambient to 25,000 psi at 210 °C, per AWES 3362-36            |
| Vibration Test                           | 3 axes, 1 hour at 10 g RMS, per AWES 3362-36                               |
| Mechanical Shock Test                    | 4 shocks in 6 directions at 500 g, 0.5 ms duration, per AWES 3362-36       |
| Thermal Cycling – Robustness Test        | Power cycling at 2 °C, –40 °C, –25 °C, and 210 °C; 50 cycles from 4–210 °C |
| Hot Transverse Shock Test                | 12,000 shocks at 210 °C, 250 g, 2 ms duration                              |
| Vertical Shock Test                      | 10 shocks in each vertical axis at 250 g, 2 ms duration                    |
| Extended HPHT Envelope – Short Cycle     | 10 cycles from ambient to 25,000 psi and 210 °C                            |
| Extended HPHT Envelope – Endurance Cycle | 40 cycles from ambient to 25,000 psi and 210 °C                            |

**Table 2—The Gauge Specifications**

| No. | Description                         | Specification                                                                      |
|-----|-------------------------------------|------------------------------------------------------------------------------------|
| 1   | Pressure Range                      | 0–10,000 psi                                                                       |
| 2   | Temperature Range                   | 25 to 177 °C [77 to 350 °F]                                                        |
| 3   | Pressure Accuracy                   | ±1.5 psi                                                                           |
| 4   | Pressure Resolution (1-s gate time) | <0.006 psi (based on Quartzdyne sensor)                                            |
| 5   | Pressure Stability / Max Drift      | ≤2 psi/year at max temperature (Quartzdyne spec)                                   |
| 6   | Temperature Accuracy                | ±0.15 °C                                                                           |
| 7   | Temperature Resolution              | 0.005 °C                                                                           |
| 8   | Temperature Stability / Max Drift   | ±0.1 °C/year                                                                       |
| 9   | Vibration and Shock Qualification   | Vibration: Per AWES 3362-36 Shock: Per AWES 3362-36 + 20 additional shocks         |
| 10  | Maximum Installation Depth          | 40,000 ft                                                                          |
| 11  | Maximum Gauges per Line             | Up to 8 dual-sensor (tubing and annulus) gauges, i.e. 16 sensors per electric line |
| 12  | Scanning Rate                       | 1-second scan rate for pressure and temperature data from all sensors              |

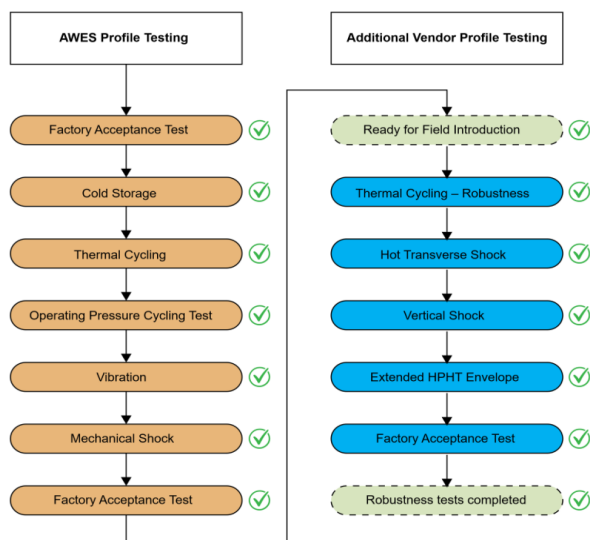


Figure 2—Gauges Validation and Verification Tests

## Well (A) Installation and Completion Schematic

Well (A) was completed as a single-lateral horizontal oil producer. The completion design incorporated 4-1/2" × 3-1/2" tubing, a subsurface safety valve (SSSV), an ESP, y-tool, and two of the newly developed gauges.

The gauges were strategically installed below the ESP and on the 3-1/2" tubing section, with both gauges monitoring the wellbore annulus. This arrangement enabled high-resolution surveillance of downhole conditions for performance evaluation. The well trajectory and final completion schematic are presented in Fig. 3.

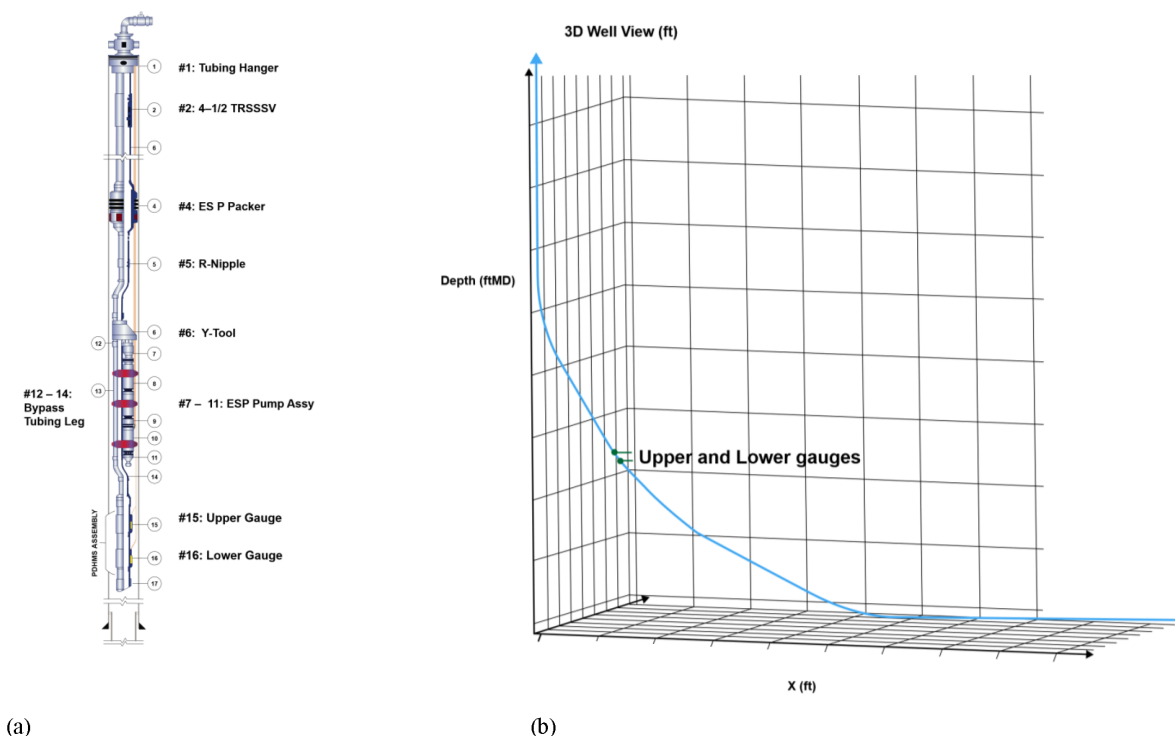


Figure 3—Well (A) Final Well Completion Schematic and Well Trajectory



## Success Criteria

The two gauges—installed at the upper and lower positions—were deployed for Well (A). The following criteria were defined to evaluate the success of technology:

1. Successful deployment to total depth (TD)
2. Full functionality of both gauges at the time of rig release
3. Either continuous operation of both gauges after three months of production, or
4. Continuous operation six months after installation in the event of a delayed production startup

## Installation, ESP Function Test and Short flowback – Gauges Readings

During the completion phase, following the landing of the tubing hanger and prior to setting the ESP packer, the well was lined up to the Tubing–Casing Annulus (TCA). Subsequently, the TCA was bullheaded with 450 bbl of 70 lbm/ft<sup>3</sup> inhibited brine at a surface pressure of 300 psi and a rate of 3 bpm.

The installed gauges successfully detected and recorded the corresponding pressure increase and temperature decline associated with the bullheading operation, as illustrated in Fig 4.

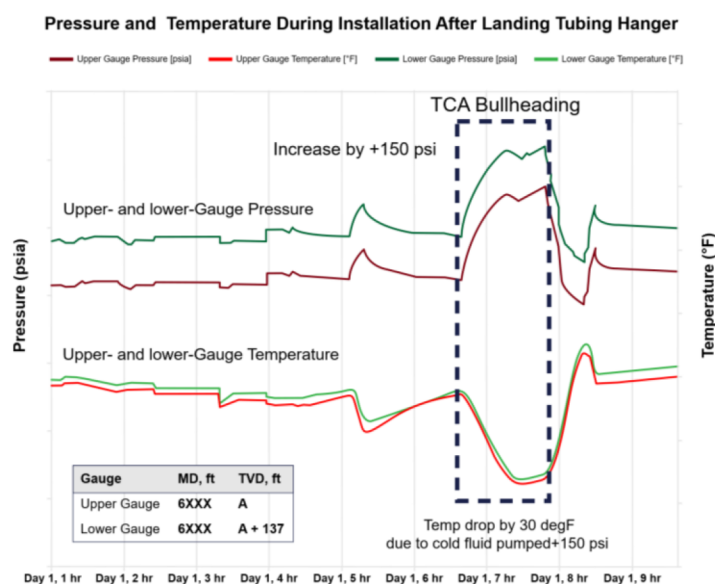


Figure 4—Gauges pressure and temperature during installation after landing tubing hanger

## ESP Function Test and Gauge Performance

Following the successful installation to total depth, landing of the tubing hanger, and setting of the ESP packer, the well was briefly flowed back to function test the ESP and assessed whether ESP-generated noise would affect gauge telemetry. The key steps of the ESP function test were:

- Initial shut-in wellhead pressure (SIWHP): 1100 psi
- ESP test conducted on a 24/64-inch choke, flowing ~80 bbl through the test setup until hydrocarbon traces were observed
- ESP shut down and surface wellhead choke closed, followed by a 1-hour monitoring period using both downhole gauges and surface wellhead gauges
- Wellhead buildup pressure: returned to 1100 psi

As shown in Fig. 5a and 5b, the gauges successfully captured the transient pressure and temperature responses during the ESP function test. During flowback, the gauges recorded a decline in wellbore pressure,

consistent with reservoir drawdown, accompanied by a temperature increase. Upon shut-in, the gauges measured the expected pressure buildup and temperature decline over the one-hour period.

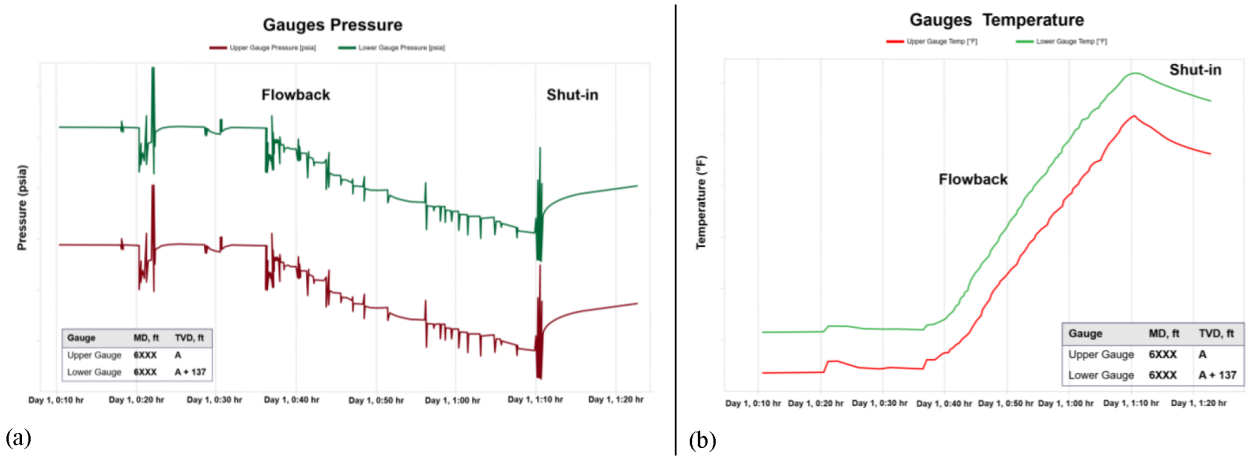


Figure 5—Gauges pressure and temperature during transient flowback conditions while functioning the ESP

Importantly, The data confirmed that ESP noise did not interfere with gauge telemetry, thereby validating the robustness of the FDMA communication system under artificial lift operating conditions.

### Six-Month Monitoring – Gauge Diagnostics, Pressure, and Temperature Measurements

Periodic well-site visits were arranged to retrieve downhole gauge data over the six-month duration following the successful installation. Gauge readings were acquired at 2-, 3-, 4-, and 6-month post-installation.

Data were recorded during each visit for 2–3 hours using a portable surface panel. The gauges demonstrated exceptional stability, capturing pressure and temperature variations as small as 0.005 psi and 0.005 °C, respectively, during shut-in conditions as shown in Fig. 6.

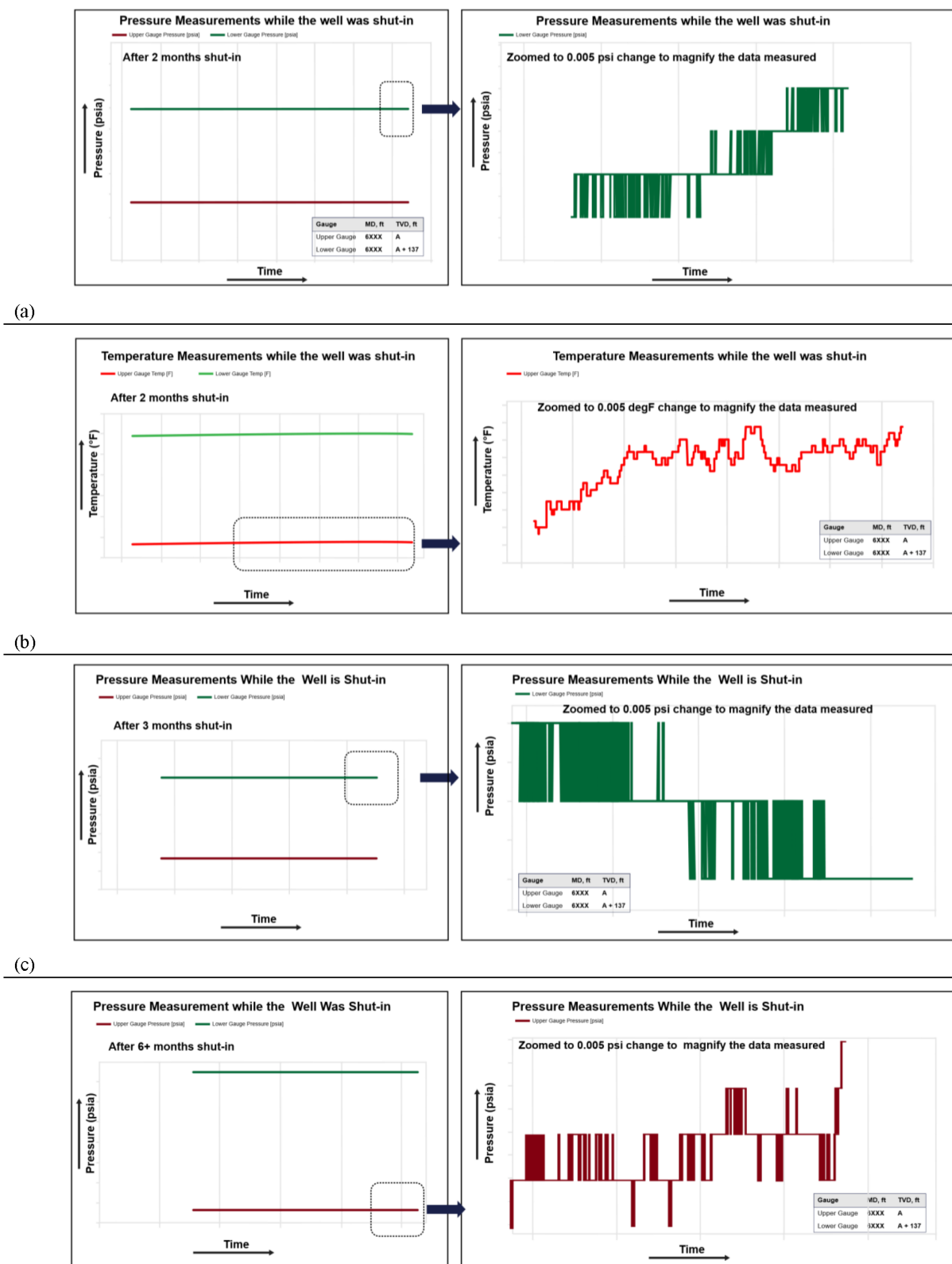


Figure 6—Gauges pressure and temperature during well shut-in conditions, 2, 3 and 6+ months after installation and shut-in

Diagnostics from the surface panel consistently indicated a green operational status, with surface output power and voltage showing no variance when compared to baseline installation readings. This confirms that both gauges remained in healthy operating condition and continued to perform reliably over the

monitoring period as demonstrated in Fig. 7. The pressure and temperature data were processed, analyzed, and visualized using dedicated software.

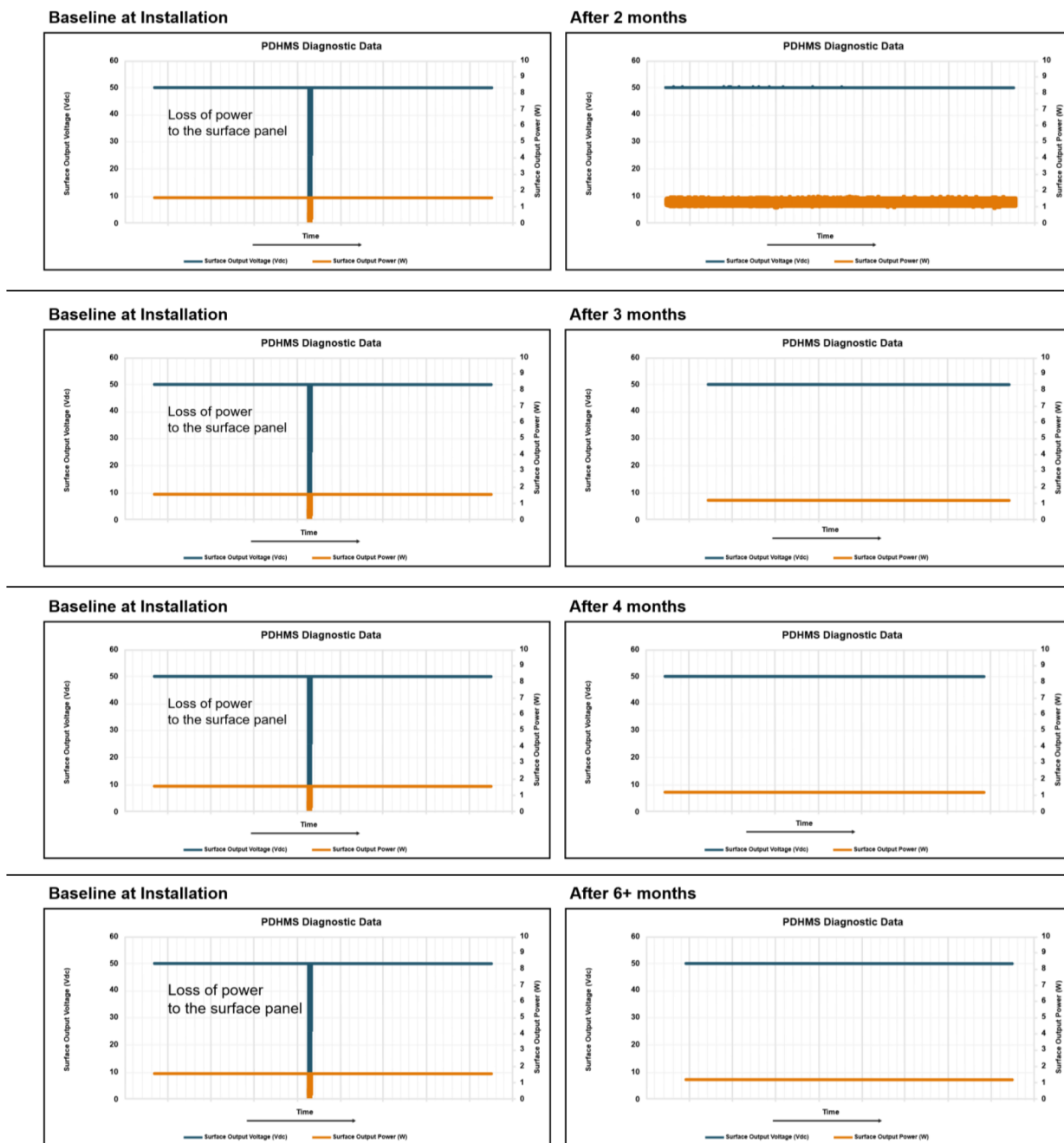


Figure 7—Gauges surface panel diagnostic and health check data during well shut-in conditions, 2, 3, 4 and 6+ months after installation

## Reservoir Insights from Six Months of Gauge Monitoring

As documented in several studies, the deployment of permanent downhole gauges in oil and gas wells has proven to be a valuable approach for continuous monitoring, enabling detailed tracking of wellbore conditions and dynamic reservoir behavior.

Analysis of the six-month dataset, acquired during the shut-in period and prior to flowback, yielded valuable insights into reservoir behavior and fluid dynamics.

- **Reservoir Pressure:** The shut-in reservoir pressure exhibited an initial decline within the first three months, followed by a +125-psi buildup between three and six months, suggesting effective pressure support from nearby water injection.
- **Reservoir Temperature:** A gradual temperature increase was observed throughout the monitoring period.
- **Fluid Gradient Evolution:**

$$\text{Fluid Gradient} = (\text{Lower Gauge Pressure} - \text{Upper Gauge Pressure}) / (\text{TVD Difference between Lower and Upper Gauge}) \quad \text{Eq. 1}$$

Gauge measurements revealed a change in fluid gradient (Eq. 1) from 0.47 psi/ft (70 lbm/ft<sup>3</sup>)—consistent with inhibited brine used during installation—to a stabilized 0.31 psi/ft, corresponding to crude oil density. This indicates a transition from brine to oil in the wellbore around the gauges, as shown on Fig. 8. These observations confirm the gauges' ability not only to deliver high-resolution, stable measurements but also to provide early diagnostic insights into reservoir dynamics and fluid movement before production startup.

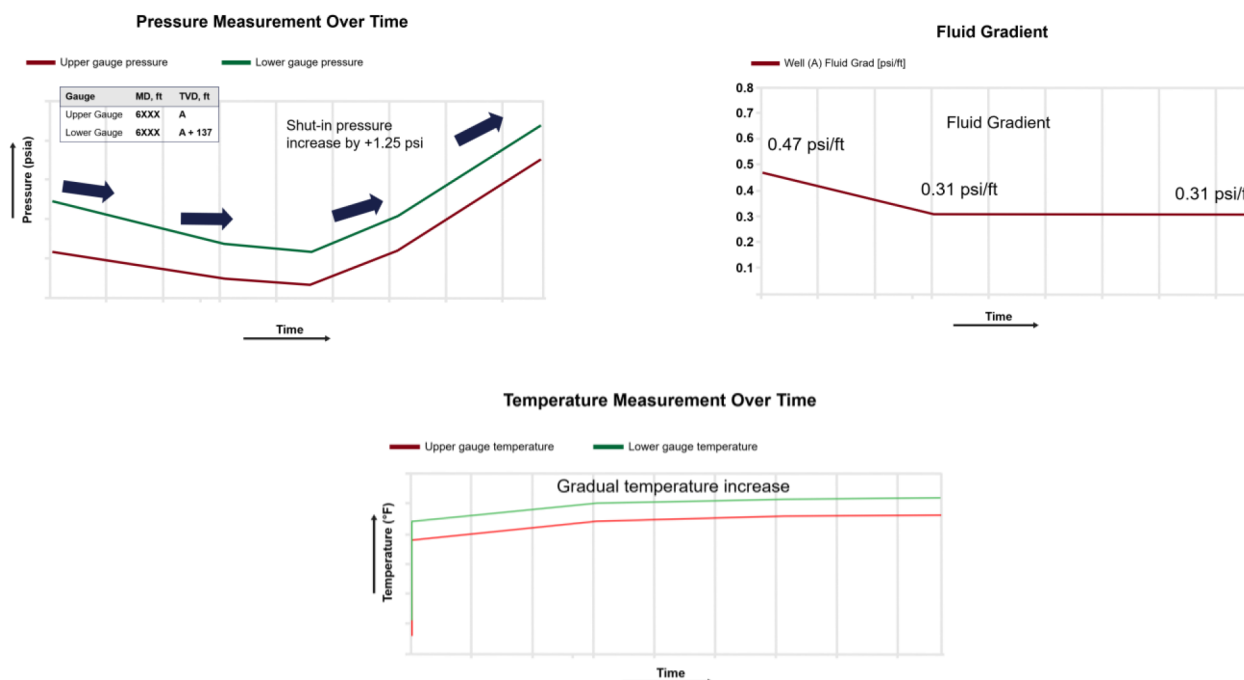


Figure 8—Reservoir insights: Comparing the gauges measurements trend for over 6 months during well shut-in conditions

## Conclusions

Based on the evaluation of the novel gauge diagnostics and the pressure and temperature measurements acquired during both the ESP function test and six months of shut-in monitoring in Well (A), several key conclusions can be drawn. First, all success criteria were achieved: both gauges were successfully deployed to total depth (TD), full functionality was verified at rig release, and the system sustained uninterrupted operation six months post-installation, despite the delayed production startup.

The gauges demonstrated reliable long-term performance. Output voltage and power diagnostics remained stable throughout the monitoring period, confirming consistent gauge health and robust downhole reliability. Notably, the system exhibited excellent immunity to electromagnetic interference. During the ESP function test, the gauges accurately captured transient downhole pressure and temperature responses, with no signal degradation from ESP-related noise—thereby validating the robustness of the FDMA telemetry protocol.

High-resolution data recorded during the extended shut-in period revealed precise pressure and temperature variations as small as 0.005 psi and 0.005 °C, underscoring the gauges' exceptional sensitivity and resolution. Furthermore, a significant transition in the fluid gradient was observed—from 0.47 psi/ft (equivalent to inhibited brine at ~70 lbm/ft<sup>3</sup>) to a stabilized 0.31 psi/ft (crude oil), providing clear evidence of fluid segregation and oil displacement around the PDHMS.

Lastly, a measured reservoir pressure buildup of 125 psi between months 3 and 6 of the shut-in period indicated active reservoir support from nearby water injection wells, demonstrating the system's value in reservoir surveillance and dynamic behavior interpretation.

## Nomenclature

|       |                                              |
|-------|----------------------------------------------|
| AC    | Alternating Current                          |
| AWES  | Advanced Well Equipment Standards            |
| CDMA  | Code Division Multiple Access                |
| DAS   | Distributed Acoustic Sensing                 |
| DC    | Direct Current                               |
| MCM   | Enhanced Multi-Chip Module                   |
| EMI   | Electromagnetic Interference                 |
| EOR   | Enhanced Oil Recovery                        |
| ESPs  | Electric Submersible Pumps                   |
| FDMA  | Frequency Division Multiple Access telemetry |
| FBG   | Fiber Bragg Gratings                         |
| FSK   | Frequency Shift Keying                       |
| HPHT  | High-Pressure, High-Temperature              |
| PDHMS | Permanent Downhole Monitoring Systems        |
| SIWHP | Shut-In Wellhead Pressure                    |
| SSSV  | Sub Surface Safety Valve                     |
| TDMA  | Time Division Multiple Access                |
| TCA   | Tubing–Casing Annulus                        |
| TD    | Total Depth                                  |

## References

- Alaseri, G., Bukhamsin, A., Garioch, A., and Abdulghani, A. 2023. Efficient Deployment of Distributed Temperature and Acoustic Sensing of Three Wells Using Fiberline Intervention Technology. Paper presented at the SPE Annual Technical Conference and Exhibition, San Antonio, Texas, USA, 16–18 October. SPE-215096-MS. <https://doi.org/10.2118/215096-MS>.
- Al-Omar, A. 2009. An Innovative Multi-Reservoir Permanent Downhole Monitoring System Through A Single Well. Paper presented at the SPE Saudi Arabia Section Technical Symposium, Al-Khobar, Saudi Arabia, 9–11 May. SPE-126158-MS. <https://doi.org/10.2118/126158-MS>.
- Antunes, P., Gonzalez, F., Yamachita, R., Nascimento, A., & Goncalves, J. L. 2015. A Review of Telemetry Data Transmission in Unconventional Petroleum Environments Focused on Information Density and Reliability. *Journal of Software Engineering and Applications* 8 (9): 455–462. <https://doi.org/10.4236/jsea.2015.89043>.
- Al Yateem, K. S., and Al Omairen, K. I. 2012. Optimization of Cableless Technologies to Obtain Reservoir Pressure and Temperature for Real-Time Monitoring. Paper presented at the SPE Intelligent Energy International, Utrecht, The Netherlands, 27–29 March. SPE-150013-MS. <https://doi.org/10.2118/150013-MS>.
- Bezerra, M. F. C., and Da Silva, S. F. 1992. Permanent Downhole Gauges: A Key To Optimize Deepsea Production. Paper presented at the Offshore Technology Conference, Houston, Texas, 4–7 May. OTC-6991-MS. <https://doi.org/10.4043/6991-MS>.
- Bouldin, B., AlShmakhy, A., Bazuhair, A. K., & Fellinghaug, J. A. 2021. A Review of Downhole Wireless Technologies and Improvements. Paper presented at ADIPEC, Abu Dhabi, UAE, 15–18 November. SPE-207466-MS.
- Mei, K. 2015. Extending Values of a PDHMS System to Production Optimization. Paper presented at the SPE Middle East Oil & Gas Show and Conference, Manama, Bahrain, 8–11 March. SPE-172738-MS.



- Mijarez-Castro Rito, R.-R., Pascasio-Maldonado, D., & Guevara-Gordillo, R. 2013. Communications System for Down-Hole Measurements. *Journal of Applied Research and Technology*, Vol. **11** Issue 6, pp. 903–911. DOI:[10.1016/S1665-6423\(13\)71597-9](https://doi.org/10.1016/S1665-6423(13)71597-9)
- NACE International. 2001. NACE MR0175: Petroleum and Natural Gas Industries—Materials for Use in H<sub>2</sub>S-Containing Environments in Oil and Gas Production. Houston, Texas, USA.
- Sadah, H., Kadem, M., Yateem, K., and Dabbous, M. 2020. Permanent Downhole Monitoring System PDHMS Reliability. Paper presented at the International Petroleum Technology Conference, Dhahran, Saudi Arabia, 13–15 January. IPTC-19634-MS. <https://doi.org/10.2523/IPTC-19634-MS>.
- SLB. 2022. Metris Extreme HPHT Permanent Pressure and Temperature Gauge Datasheet. SLB, Houston, Texas, USA.
- Zeghlache, L., Bukhamseen, A., Muailu, H., and Abdulghani, A. 2022. Sensor-Ball: Field Deployment of Autonomous and Untethered Surveillance. Paper presented at the International Petroleum Technology Conference, Riyadh, Saudi Arabia, 21–23 February. IPTC-22255-MS. <https://doi.org/10.2523/IPTC-22255-MS>.